

Mahmoud Alyosify

Ai & Data Science Engineer | MSc (Ai) @Queen's University
First Section, New Cairo, Egypt (Easily Relocate)

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Kaggle: <https://www.kaggle.com/mahmoudalyosify> | HuggingFace: <https://huggingface.co/mahmoudalyosify>

SUMMARY

AI and Data Science Engineer with a strong foundation in machine learning, data engineering, and cloud-based AI deployment. Currently pursuing an MSc in Artificial Intelligence through the competitive Digilians Scholarship in partnership with Queen's University, Canada (January 2026 – Present). Proven track record delivering end-to-end AI solutions — from data pipelines and model training to production APIs on AWS. Strong software engineering background with full-stack (.NET) development experience. Also run the "[Egypt's Einstein](#)" educational YouTube channel on computer science topics (10.4K+ subscribers, 716K+ views). Delivered technical education to 10,000+ students online ([Udemy](#)). Passionate about Quantum ML, automation, data science, and building real-world AI solutions for enterprise clients.

EDUCATION

- ❖ **MSc in Artificial Intelligence (In Progress, Jan 2026 – Present) | Queen's University, Canada**
 - Within the scholarship of the President of the Arab Republic of Egypt (Digilians).
 - **Completed Coursework:** CSAI 801 — Fundamentals of AI & ML | CSAI 810 — Topics in AI (Generative AI, LLMs, RAG, AI Agents) | CISC 886 — Cloud Computing & Big Data (Spark, Hadoop, Docker, CI/CD, AWS) | CISC 839 — Advanced Data Analytics (End-to-End ML Pipelines, GenAI for Analytics).
 - **Currently Taking:** CISC 867 — Deep Learning | CISC 856 — Reinforcement Learning.
- ❖ **Assiut University — B.Sc. Computer & Information Science (Bioinformatics) | Graduated July 2023**
 - **GPA:** 3.53 | **Graduation Project:** [Vitalism Solution](#) — AI system for contactless estimation of vital signs via video (rPPG + wavelet filtering). Grade: A+

TECHNICAL SKILLS

AI / ML: TensorFlow, PyTorch, Scikit-learn, XGBoost, CNNs, NLP, OpenCV, NLTK, Librosa

Data Engineering: Apache Kafka, Pandas, NumPy, Feature Engineering, ETL Pipelines, Hadoop

Cloud & MLOps: AWS Lambda, Docker, Git/GitHub, FastAPI

Languages: Python, C#, R, JavaScript, Go, C++

Backend / DB: ASP.NET Core, Django, Flask, SQL Server, MongoDB, Entity Framework

Frontend: Angular, HTML/CSS, Bootstrap, Streamlit

WORK EXPERIENCE

July 2025 – September 2025

- ❖ **NTI - National Telecommunication Institute, [Machine Learning Instructor](#)**
 - Delivered a 90-hour hands-on course covering supervised & unsupervised learning (LR, SVM, KNN, RF, PCA, NN) with real-world projects, training engineering & CS students on building end-to-end ML solutions.

June 2025 – July 2025

- ❖ **e& Egypt (via National Telecommunication Institute OJT) , [AI in Cybersecurity Intern](#)**
 - Covered cyber fundamentals with focus on threat intelligence and hands-on tools (Kaspersky, ISE, DLP, Volatility).

February 2024 – March 2025

- ❖ **National Company for Roads Building & Development, [Systems Engineer \(Military Service\)](#)**
 - Maintained mission-critical backend systems powering Egyptian toll stations and smart parking systems, serving thousands of cars daily (covering over 50% of Egyptian vehicles across 70% of the roads).
 - Co-developed a multi-department data management desktop system using C#, .NET, and SQL Server.
 - Performed advanced troubleshooting, debugging, and system optimization to ensure high availability and performance of critical applications, utilizing C#, .NET, Entity Framework, SQL Server, and Crystal Reports.

2020 – Present

- ❖ **Udemy + YouTube, [Online Instructor \(AI, Algorithms & Data Science\)](#)**
 - Delivered Arabic-language courses to 10,000+ enrolled students covering Discrete Math, Algorithms, Graph Data Mining, and Probability.
 - Built a YouTube channel with 700,000+ views — communicating complex CS concepts to diverse audiences.

KEY PROJECTS

- ❖ [SimCLR-Vision-SSL — Self-Supervised Contrastive Learning](#) | **May 2026 – Present**
 - **In Progress** • PyTorch · SimCLR · BYOL · CIFAR-10 · STL-10 · t-SNE
 - Implementing SimCLR self-supervised contrastive learning framework on CIFAR-10 and STL-10 datasets — learning rich visual representations without labelled data.
 - Evaluation pipeline includes linear probe, KNN evaluation, and BYOL comparison; results visualised via t-SNE to interpret learned feature space.
- ❖ [Horus-OSINT — Cloud-Based Threat Intelligence Assistant](#) | **Mar 2026 – May 2026**
 - **Queen's University, Canada** • Python · Meta-Llama-3-8B · AWS · LLM Fine-Tuning · GTD & GDELT Datasets
 - Fine-tuned Meta-Llama-3-8B-Instruct on 159,826 records from the Global Terrorism Database (GTD) and GDELT datasets to build a production-grade OSINT conversational threat analyst.
 - Deployed as a cloud-native web application on AWS, enabling real-time global threat intelligence queries through a fully interactive chat interface.
- ❖ [Analyzing Review Effort in Human vs. Agentic Pull Requests](#) | **Mar 2026 – May 2026**
 - **Queen's University, Canada** • Investigated differences in review effort between AI-generated (agentic) and human-authored pull requests in highly active open-source GitHub repositories using a full analytical replication pipeline and data-driven evaluation.
- ❖ [SCRAP — Satellite Collision Risk Assessment & Prediction](#) | **Jan 2026 – Mar 2026**
 - **Queen's University, Canada** • Python · Scikit-learn · Supervised ML · Telemetry Data · LEO Orbital Mechanics
 - Built a supervised ML framework to predict satellite-debris collision risk in Low Earth Orbit using only telemetry data available 48+ hours before Time of Closest Approach — matching ESA's hard operational deadline.
 - Significantly reduced false positives in a high-noise alert environment where fewer than 1% of hundreds of weekly ESA collision warnings require real action.
- ❖ [Automated Multimodal Agent — PDF to Narrated PowerPoint](#) | **Jan 2026 – Mar 2026**
 - **Queen's University, Canada** • Python · Hugging Face · Generative AI · NLP · Agentic AI · TTS
 - Designed an autonomous agent-based system (Think–Plan–Act paradigm) that converts any PDF into a fully narrated PowerPoint presentation without human intervention.
 - Engineered a workaround for python-pptx audio limitations by converting TTS narration to MP4 and embedding as invisible auto-playing media within each slide.
- ❖ [AudioShield — Deepfake Audio Detection System](#) | **Apr 2025 – Jun 2025**
 - **NTI** • Python · TensorFlow · Librosa · CNN · Streamlit
 - Built a CNN-based deepfake audio detector using mel-spectrogram feature extraction — targeting fraud detection in call centers, financial services, and virtual meetings.
 - Deployed as an interactive Streamlit application with real-time prediction and automated PDF reporting.
- ❖ [E-Commerce Web Application](#) | **Angular · Bootstrap · Stripe API · JWT Authentication**
 - **Route Academy** • Interactive e-commerce platform with product listing, cart, and secure auth; integrated Stripe API for payments and Angular Routing for navigation.
- ❖ [Vitalism Solution — Contactless Vital Signs via Video](#) | **Jun 2022 – Jul 2023** | **Grade: A+**
 - **Assiut University — Graduation Project** • Python · rPPG · Wavelet Filtering · OpenCV
 - AI system extracting HR, HRV, SpO2, respiratory rate, and blood pressure from standard video streams — no physical sensors required; targets telemedicine and low-resource healthcare.
- ❖ [Diabetes Onset Prediction](#) | **Python · Pandas · Scikit-learn · Machine Learning**
 - Built a predictive model to estimate the risk of diabetes based on diagnostic and metabolic features (e.g., BMI, insulin, age, pregnancies), leveraging classification algorithms to identify early risk patterns from medical data.
- ❖ [ATM Transaction Data Analysis](#) | **R · dplyr · ggplot2 · SQL**
 - **Children's Cancer Hospital Egypt 57357** • Analysed ATM availability, downtime, and manufacturer reliability (NCR vs. Diebold Nixdorf); identified key performance trends using R and SQL.
- ❖ [USCB Data Analysis](#) | **R · dplyr · ggplot2 · SQL**
 - **Children's Cancer Hospital Egypt 57357** • Performed statistical analysis on US Census Bureau data to uncover population trends, income distribution, and economic patterns; visualised gender equality, education levels, and income disparities across demographics.
- ❖ [Mini-FAT File System & Shell](#) | **C++ · Operating Systems**
 - Engineered a FAT-based file system with a custom shell supporting cd, copy, del, and import/export commands; designed for efficiency and robustness.
- ❖ [Student Affairs System](#) | **C# · Windows Forms · SQL Server**
 - Desktop student management system tracking academic records, analysing performance, and automating report generation.

CERTIFICATIONS

- ❖ **AWS Academy Graduate — Data Engineering** — Amazon Web Services | 2026
- ❖ **AWS Academy Graduate — Machine Learning for NLP** — Amazon Web Services | 2025
- ❖ **AWS Academy Graduate — Machine Learning Foundations** — Amazon Web Services | 2025
- ❖ **Machine Learning Specialization** — Coursera / DeepLearning.AI (Andrew Ng)
- ❖ **Mathematics for Machine Learning** — Coursera
- ❖ **HCIP — AI (Huawei Certified ICT Professional)** — Huawei
- ❖ **Microsoft Certified: Data Analyst Associate** — Microsoft (Power BI)
- ❖ **Generative AI & Prompt Engineering Essentials** — IBM / Coursera
- ❖ **Generative AI: Boost Your Cybersecurity Career** — IBM
- ❖ **AI in Cybersecurity** — NTI | 420 hours (Mar 2025 – Jul 2025)
- ❖ **Machine Learning Internship** — ITIDA / Egypt Makes Electronics | 200 hours (Jul – Sep 2023)
- ❖ **Full Stack Development Diploma (.NET & Angular)** — Route Academy | May 2023 – Feb 2024
- ❖ **Data Analysis Training using R** — Children's Cancer Hospital Foundation 57357
- ❖ **.NET Web Development (120 hrs.)** — ITI (ASP.NET MVC, Web API, Razor, Blazor, SignalR, Identity)
- ❖ **SQL Server & C# (90 hrs.)** — ITI (DB design, stored procedures, LINQ, Entity Framework)
- ❖ **MERN Stack Development (60 hrs.)** — ITI (MongoDB, Express.js, React.js, Node.js)
- ❖ **UI/UX Design (60 hrs.)** — ITI (UX fundamentals, design process, prototyping)
- ❖ **Android Mobile Development (30 hrs.)** — ITI (Android Studio, Java, XML)

ACHIEVEMENTS & AWARDS

- ❖ **1st Place — Smart Cities Hackathon 2022** (competed vs. governmental, private & national universities in Egypt)
- ❖ **Semi-Finalist — Microsoft Imagine Cup 2023** (Vitalism Project qualified for global semi-finals)
- ❖ **Finalist — ISEIC 2023** (International AI Research Competition — competed vs. 800+ teams at Air Defense College)
- ❖ **Ideal Student Award** — Faculty of Computers & Information, 2022–2023
- ❖ **1st Place — Science & Technology Content Competition** — White Hackers Team (University level, 2021–2022)

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- ❖ **Secretary — Higher Scientific & Technological Committee, Student Union | Assiut University, 2021–2022**
 - Encouraged students to participate in scientific and technological competitions, inventions, and innovations; acted as a bridge between students and university administration.
 - Honoured by external institutions for outstanding efforts in making student activities successful.
- ❖ **Ambassador — Institute of Information Technology (ITI) in Assiut**
 - Represented ITI as a student ambassador, promoting technology education and career development opportunities.
- ❖ **Volunteer Team Leader — University Center for Career Development (USAID Partnership) | Assiut University**
 - Bridged the gap between university education and labor market requirements through structured volunteer initiatives.
- ❖ **Board Member — Smile Makers Association for Orphan Sponsorship & Community Development | Assiut**
 - Volunteered in the summer educational club; taught mathematics, science, and programming to children to develop their scientific skills.

ACTIVITIES PROFESSIONAL DEVELOPMENT

- ❖ **Employability Skills** — Aspire in cooperation with UCCD, Assiut University
- ❖ **Project Management** — American Chamber of Commerce in Egypt
- ❖ **Green & Smart Technologies Boot Camp** — British University in Egypt
- ❖ **Leaders Preparation Camp.** — Assiut University

LANGUAGES

- ❖ **Arabic** — Native
- ❖ **English** — Good

HOBBIES

- ❖ **Fish Hunting**
- ❖ **Photography**
- ❖ **Volleyball**
- ❖ **Cooking**